

- Find the terminal velocity of a spherical particle (radius $1 \mu\text{m}$, density 1050 kg/m^3) falling in water ($\rho_{\text{water}} = 1000 \text{ kg/m}^3$). [Hint. Terminal velocity is the velocity of the particle where drag force and gravitation force are balanced.] (3)
- A particle experiences a drag force of 5 pN while moving at a velocity of $2 \mu\text{m/s}$ in a fluid. Find the viscous friction coefficient. (2)
- A sphere of radius 2 mm is moving through water at 27°C . Find out its diffusion coefficient. (3)
- Identical microscopic particles are diffusing in a fluid kept in a 500 mL beaker. If the RMS displacement of the particles after 10 minutes is 1.6 mm , find out the diffusion coefficient of the particles. (2)
- Thoughtful Question: Pointwise and with the help of equations, solve the problems given in the table with both quantitative and qualitative (comments) answers. (5)

Given:

Bacterium (E. coli)	Human
$L \approx 2 \mu\text{m}$	$L \approx 1.5 \text{ m}$
$v \approx 20 \mu\text{m/s}$	$v \approx 1 \text{ m/s}$
$\rho_{\text{water}} = 1000 \text{ kg/m}^3$, $\eta_{\text{water}} = 0.001 \text{ kg/(m.s)}$	Same fluid properties

Find out the differences in a table as shown below:

Aspect	Bacteria	Human	Comments
R_e	< 0.005	2×10^6	
Drag force	30 fN	$\sim 16 \text{ N}$	
Propulsion	Non-reciprocal	Reciprocal	
Reciprocity			
Power per mass	$\sim 6/7 \times 10^{-18} \text{ W}$ $600-700 \text{ zW}$	6.33 W/kg	

$m = 4 \times 10^{-18} \text{ kg}$

Or, $1.6 \times 10^{-18} \text{ W/kg}$

- Describe using Ray-optics diagrams and equations the trapping of particles (for (a) $\lambda < d$; (b) $\lambda > d$) by Photons. (5)

$\rightarrow \gamma \propto \sqrt{V}^2 A$
 $L \rightarrow \text{Area}$
 $|\mathbf{f}| = \eta |\dot{\mathbf{V}}| A$

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①. $6\pi\eta r v = mg$
 $v = \frac{mg}{6\pi\eta r} = \frac{\rho_0 V_0 g}{6\pi\eta r} = \frac{1050 \times \frac{4}{3} \pi \times 10^{-6} \times 9.8}{6 \times \pi \times 0.001 \times 10^{-6}}$
 $= \frac{50 \times \frac{4}{3} \times 10^{-12}}{6}$

$\frac{4}{3} \pi r^3 \rho_0 g + 6\pi\eta r v = \frac{4}{3} \pi r^3 \rho g$

$v = \frac{2r^2(\rho - \rho_0)g}{9\eta}$

$= \frac{2 \times (10^{-6})^2 (50) \times 10}{9 \times 10^{-3} \text{ Pa.s}}$

$= \frac{10^3 \times 10^3 \times 10^{-12}}{9}$

$= 10^{-6}/9 = 0.11 \times 10^{-6} \text{ m/s}$

$\rho = m/v$

$F = mg$
 $F = 6\pi\eta r v$

$\eta = 10^{-3} \text{ Pa.s}$

$\frac{\text{pN}}{\mu\text{m}} \times \frac{\text{m} \times \text{m/s}^2}{\text{m} \times \text{m}} \times \frac{\text{kg}}{\text{kg}}$